Open problems:
Non-linear representations in quantum mechanics

Open problems raised by the technical brief in this directory, and by its Lean formalization (see `formalization-log.tex`). Items marked [BLOCKS VERIFICATION] stand in the way of checking a main claim of the brief as the brief states it: a *formalization gap* can be closed in Lean alone, while a *brief issue* needs the brief to be corrected or made precise first.

From the brief

1. **Transport of the rest of quantum mechanics.** The brief claims that "every other relationship in quantum mechanics can be expressed in this non-linear space through m", including superpositions, expectations of observables and time evolution. Only rays and the Born rule are checked. (Brief, Sec. 3, Remark.)

2. **No benefit from non-linear representations.** The transported relationships "would become considerably more complicated for likely no benefit". This is a speculation; no criterion for "benefit" is given. (Brief, Sec. 3, Remark.)

3. **Madelung representation.** "The Madelung equations ... provide such a representation", one "not even tied to vector spaces". The equivalence of the Madelung equations with quantum theory, and in what sense they represent QST, are not shown. (Brief, Sec. 4.)

4. **Any dimension.** The proof needs a non-linear, Born-rule-preserving, ray-preserving map on the $\mathcal{H}$ of the system, but constructs m only for a two-state system. The same formula $m(\psi) = e^{\imath\theta\, p(z^+|\psi)}\psi$ for a unit vector $|z^+\rangle$ should work whenever $\dim \mathcal{H} \geq 2$, including infinite dimension; this is not formalized ($\mathcal{H} = \mathbb{C}^2$ only). (Brief, Sec. 3, eq. (2); Log, "Setting".)

From the formalization

1. [BLOCKS VERIFICATION: BRIEF ISSUE] **Theorem 1 as an independence statement.** "REP-VSLIN is independent of QST" is a statement about logical independence; only the existence of a non-additive homeomorphism of $\mathcal{H}$ that fixes rays and preserves the Born rule is proved. A precise formulation of ¬REP-VSLIN (the map m sends every ray to itself, so $m(\mathcal{H}) = \mathcal{H}$ as a set) under which the brief's inference holds is not settled. (Log, "Differences from the brief".)